

ARVIN ARYANPOUR

Software Engineer

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SUMMARY

Software engineer with full-stack, computer-vision, and ML experience spanning hardware, backend, and frontend development. Experienced in building scalable systems from real-time computer vision pipelines and microservices architectures to polished user interfaces. Strong foundation in system design, object-oriented principles, and Agile workflows, with a track record of translating complex technical requirements into efficient, maintainable, and user-focused solutions. Passionate about working across the stack and contributing to teams that value clean code, iterative improvement, and measurable impact.

SKILLS

Programming: Python, C/C++, Java, JavaScript, C#, SQL, VHDL, MATLAB

AI/ML & Computer Vision: PyTorch, YOLOv8, OpenCV, ByteTrack, scikit-learn, NumPy, Reinforcement Learning (PPO, Unity ML-Agents)

Backend & Systems: FastAPI, REST APIs, microservices, PostgreSQL, SQLite, WebSockets

Frontend: React, HTML/CSS, UI/UX integration

Tools & Platforms: Git, Linux/UNIX, FPGA (Quartus), Arduino, Unity

Core Strengths: System design, embedded systems, computer vision, OOP, Agile/Scrum

EXPERIENCE

Web Developer

Pro Insulation Contracting

Dec 2024 - Jan 2026 . Toronto, ON

- Built lead-generation web applications in FastAPI with REST APIs, improving conversion rates and user engagement.
- Optimized performance and SEO, reducing load times and increasing inbound traffic.
- Integrated analytics pipelines to track user behaviour and marketing-performance metrics.
- Translated business requirements into scalable technical solutions across frontend and backend systems.

Software Engineer Intern

VerticalScope

Apr 2023 - Oct 2023 . Toronto, ON

- 5+ reusable UI components developed across 1,200+ enthusiast websites, improving frontend consistency.
- 200ms reduction in homepage load times, contributing to a measurable increase in user retention and ad impressions.
- 15% optimization in backend API response times through strategic database indexing and query refactoring.
- 65% to 80% increase in codebase test coverage, preventing critical regressions in the production environment.

Footwear Sales Associate

Sporting Life

Sep 2024 - Present . Toronto, ON

- #1 ranked in department sales and 5-star reviews, applying data-driven decision-making and analyzing customer behaviour patterns.
- Translate complex product specifications into clear, user-focused recommendations that consistently exceed KPIs.
- Adapt strategies in real time based on feedback, demonstrating iterative problem-solving in high-volume environments.

PROJECTS

- **Real-Time Volleyball Analytics Platform:** Built a YOLOv8 + ByteTrack computer-vision pipeline achieving 87.9% precision for player and ball tracking; engineered the full FastAPI + React video stack for real-time inference, frame-level analytics, and an interactive review UI.
- **Smart Parking Microservices System:** Engineered a microservices-based platform with dynamic pricing logic and authenticated REST APIs across independent services; designed for horizontal scalability with PostgreSQL persistence and clean inter-service contracts.
- **Reinforcement Learning Agent (Unity ML-Agents):** Designed and trained a Proximal Policy Optimization (PPO) agent in Unity ML-Agents, including custom reward shaping, observation-space design, and full training-loop instrumentation to validate convergence behaviour.

EDUCATION

B.Eng., Computer Software Engineering

Toronto Metropolitan University

2021 - 2026 . Toronto, ON

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Object-Oriented Analysis & Design, Software Engineering, Network Security, Embedded Systems, Computer Architecture, Computer Vision, Intelligent Systems

COUNCILS AND CLUBS**MUES Engineering Council** - Event Coordinator

Planned and coordinated faculty-wide engineering events as part of the Metropolitan University Engineering Society, fostering community engagement and professional development across programs.

TMU Formula Racing Team - Software Team

Contributed to embedded software and data systems on TMU's student-built formula race car, collaborating with a multidisciplinary team across two competitive seasons.